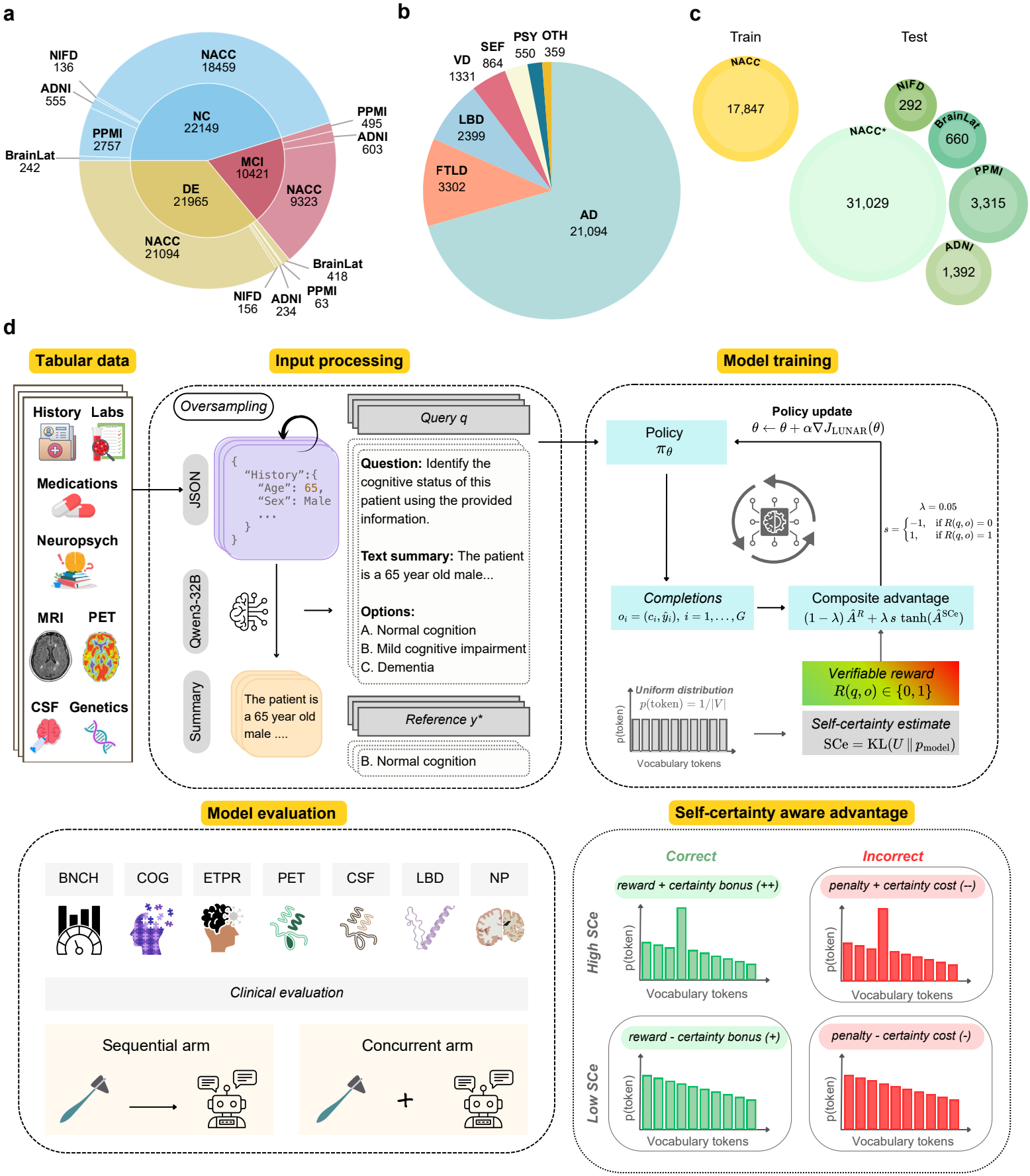

a
NIFD 136
ADNI 555
NACC 18459
NC 22149
PPMI 2757
BrainLat 242
MCI 10421
DE 21965
NACC 9323
NACC 21094
PPMI 495
ADNI 603
BrainLat 418
NIFD 156
ADNI 234
PPMI 63
b
VD 1331
SEF 864
PSY 550
OTH 359
LBD 2399
FTLD 3302
AD 21,094
c
Train
Test
NACC 17,847
NACC* 31,029
NIFD 292
BrainLat 660
PPMI 3,315
ADNI 1,392
d
Tabular data
History
Labs
Medications
Neuropsych
MRI
PET
CSF
Genetics
Input processing
Oversampling
JSON
{ "History":{ "Age": 65, "Sex": Male ... } }
Qwen3-32B
Summary
The patient is a 65 year old male ....
Query q
Question: Identify the cognitive status of this patient using the provided information.
Text summary: The patient is a 65 year old male...
Options:
A. Normal cognition
B. Mild cognitive impairment
C. Dementia
Reference y*
B. Normal cognition
Model training
Policy
π_θ
Policy update
$\theta \leftarrow \theta + \alpha \nabla J_{\mathrm{LUNAR}}(\theta)$
$\lambda = 0.05$
$s = \begin{cases} -1, & \text{if } R(q,o) = 0 \\ 1, & \text{if } R(q,o) = 1 \end{cases}$
Completions
$o_i = (c_i, \hat{y}_i),\ i = 1, \ldots, G$
Composite advantage
$(1-\lambda)\,\hat{A}^R + \lambda\, s\, \tanh(\hat{A}^{\mathrm{SCe}})$
Verifiable reward
$R(q,o) \in \{0,1\}$
Uniform distribution
$p(\text{token}) = 1/|V|$
p(token)
Vocabulary tokens
Self-certainty estimate
$\mathrm{SCe} = \mathrm{KL}(U \,||\, p_{\mathrm{model}})$
Model evaluation
BNCH
COG
ETPR
PET
CSF
LBD
NP
Clinical evaluation
Sequential arm
Concurrent arm
Self-certainty aware advantage
Correct
Incorrect
High SCe
Low SCe
reward + certainty bonus (++)
penalty + certainty cost (--)
reward - certainty bonus (+)
penalty - certainty cost (-)
p(token)
Vocabulary tokens